

Recommendation systems.

- Designed to understand users & product preferences by analyzing interaction data e.g. clicks.
- 3 Types of Recommendation Systems:
 - Content based
 - Collaborative filtering
 - Context filtering
- Recommendation systems rely on feedback i.e. implicit & explicit.
 - Explicit → User directly rates something
 - Implicit → Based on user actions. Abundant and easy to collect but is less precise and can be noisy.
- user-item rating matrix
- Content based → recommends by matching the attributes of items with user's preferences.
- Utility Matrix
- Each item in system is described by a profile that captures essential characteristics.
- Term Frequency → Count of how often a word appears relative to most frequent word.

$$TF_{ij} = \frac{f_{ij}}{\max_k f_{kj}}$$

- Inverse Document Frequency → How unique a word is across all documents.

$$IDF_i = \log \frac{N}{n_i}$$

- TF-IDF score = $TF_{ij} * IDF_i$

- Use 2 main methods for content based:

✓ Cosine similarity

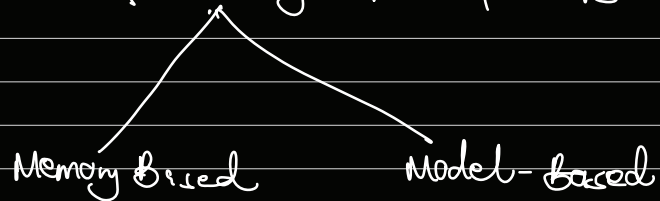
user-profile

item-profile

The smaller the better

✓ Classification models.

→ Collaborative filtering uses collective knowledge & behavior of a large pool of users.



→ Memory Based uses entire user-item matrix to make direct recommendations.